

Building Your First LangGraph Workflow

Overview

You've explored the theory behind LangGraph's architecture and seen how it works in production. Now it's time to experience the power of graph-based workflows firsthand by building your own system.

Scenario

You're a developer at a growing SaaS company that receives hundreds of support tickets daily. Currently, all tickets go into a single queue, causing delays and frustration. Your team wants to build an intelligent ticket routing system that can classify incoming tickets by urgency and topic, then route them to the appropriate support specialists.

The system needs to handle various ticket types: billing issues, technical problems, feature requests, and account management. It should also flag urgent tickets that need immediate attention and route complex technical issues to senior engineers.

Build a LangGraph workflow that processes support tickets through classification, urgency assessment, and intelligent routing to appropriate handlers.

Instructions for learners

Task 1: Define Your State Schema (3 minutes)

Create a state structure that will hold all the information needed throughout your workflow:

Original ticket content

Classification results (category and confidence)

Urgency level

Routing decision

Any additional metadata

Task 2: Design Your Workflow Nodes (4 minutes)

Plan and outline three core nodes for your system:

Classifier Node: Analyzes ticket content and determines category (billing, technical, feature, account)

Urgency Assessor Node: Evaluates ticket urgency based on keywords and context

Router Node: Makes final routing decision based on classification and urgency

For each node, define:

- What input it expects from the state
- What processing it performs
- What it adds to or updates in the state

Task 3: Map Your Conditional Logic (3 minutes)

Design the decision flow between your nodes

When should tickets go directly to routing vs. additional processing?

How should high-urgency tickets be handled differently?

What conditions trigger different routing paths?

Create a simple flowchart or written description of your conditional edges.

Completion Criteria

State schema with all necessary fields defined

Three nodes with clear input/output specifications

Conditional logic flow that handles different ticket scenarios

Brief explanation of how your workflow improves upon a simple linear process

Engage in reflection: Transition to Dialogue

Now that you've designed your first LangGraph workflow, you'll enter a guided conversation to examine your architectural decisions and explore how they align with production-ready system design principles.